

## Clarkson Institute Summer School: Examples Sheet 2 - Basic properties of group elements

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### Proving properties of abstract groups

1. Consider the equation  $x * x = x$  for the variable  $x$  in a group. Show that the unique solution of this equation is the identity of the group.

*Solution:* Suppose that  $x$  solves the equation. Then:

$$\begin{aligned}x * x = x &\quad \Rightarrow \quad x^{-1} * (x * x) = x^{-1} * x && \text{(multiplying on left by } x^{-1}) \\&\quad \Rightarrow \quad (x^{-1} * x) * x = e && \text{(associativity, inverses)} \\&\quad \Rightarrow \quad e * x = e && \text{(inverses)} \\&\quad \Rightarrow \quad x = e && \text{(identity)}\end{aligned}$$

Hence if  $x$  solves the equation,  $x = e$ . On the other hand,  $e * e = e$  by the axiom of inverses, so the identity does indeed solve the equation. It follows that the identity is the unique solution.

2. Using the group axioms, prove the following laws for group elements  $x, y$ :

- (a)  $(x^{-1})^{-1} = x$ ;  
(b)  $(xy)^{-1} = y^{-1}x^{-1}$ .

How does the second property generalise to  $(x_1x_2\dots x_n)^{-1}$ , where  $x_1, x_2, \dots, x_n$  are group elements? [You may assume the generalised associative law here, so that  $x_1x_2\dots x_n$  is unambiguous.]

*Solution:*

- (a) The axiom of inverses tells us that  $x * x^{-1} = e$ . The inverse of the element  $x^{-1}$  is unique, and must therefore be equal to  $x$ , hence  $(x^{-1})^{-1} = x$  by definition.  
(b) Observe that  $(xy)(y^{-1}x^{-1}) = x(yy^{-1})x^{-1} = xx^{-1} = e$ . Hence  $y^{-1}x^{-1}$  is the inverse of  $xy$ , i.e.  $(xy)^{-1} = y^{-1}x^{-1}$ .

The second property generalises to  $(x_1x_2\dots x_n)^{-1} = x_n^{-1}x_{n-1}^{-1}\dots x_1^{-1}$ , since:

$$(x_1x_2\dots x_n)(x_n^{-1}x_{n-1}^{-1}\dots x_1^{-1}) = e.$$

3. Let  $a_1, a_2, \dots$  be elements of a group  $G$  with operation  $*$ . Define  $S_1(a_1) = \{a_1\}$ ,  $S_2(a_1, a_2) = \{a_1 * a_2\}$ , and:

$$S_n(a_1, \dots, a_n) = \{x * y : x \in S_p(a_1, \dots, a_p), y \in S_q(a_{p+1}, \dots, a_n), p + q = n, p, q \geq 1\}.$$

- (a) Explain why the set  $S_n(a_1, \dots, a_n)$  contains all possible elements formed from inserting brackets into the product  $a_1 * a_2 * \dots * a_n$ . Hence, explain why the set  $S_3(a_1, a_2, a_3)$  contains one element.
- (b) Using induction, prove that  $S_n(a_1, \dots, a_n)$  contains only a single element. Deduce the *generalised associative law*: in a group, the product  $a_1 * a_2 * \dots * a_n$  is independent of the placement of brackets.

*Solution:*

- (a) Consider where the brackets could go around  $a_1$ . If there is no bracket around  $a_1$ , the bracketing must be of the form:

$$a_1 * (\underbrace{a_2 * \dots * a_n}_{\text{some bracketing}}),$$

which is of the form  $x * y$  with  $x \in S_1(a_1)$  and  $y \in S_{n-1}(a_2, \dots, a_n)$ . On the other hand, if there is any bracket preceding  $a_1$ , the expression takes the form:

$$(\underbrace{((\dots(a_1 * \dots * a_p))}_{\text{expression up to close of initial bracket}} * \underbrace{(a_{p+1} * \dots * a_n))}_{\text{some bracketing}}).$$

This is clearly in the set  $S_n(a_1, \dots, a_n)$  as it is of the form  $x * y$  where  $x \in S_p(a_1, \dots, a_p)$  and  $y \in S_q(a_{p+1}, \dots, a_n)$ , where  $p + q = n$ . We conclude that indeed all possible bracketings of the expression  $a_1 * a_2 * \dots * a_n$  are contained in the set  $S_n(a_1, \dots, a_n)$ .

The set  $S_3(a_1, a_2, a_3)$  only contains the bracketings  $a_1 * (a_2 * a_3)$  and  $(a_1 * a_2) * a_3$ , but these are equal by the associative law in the group.

- (b) We have seen that  $S_1(a_1)$ ,  $S_2(a_1, a_2)$  and  $S_3(a_1, a_2, a_3)$  all contain one element, so our base case is established. Now suppose that  $S_{n-1}(a_1, \dots, a_{n-1})$  contains only one element (for any choice of group elements  $a_1, \dots, a_{n-1}$ ).

Let  $x * y \in S_n(a_1, \dots, a_n)$ . Then there exist  $p, q \geq 1$  with  $x \in S_p(a_1, \dots, a_p)$  and  $y \in S_q(a_{p+1}, \dots, a_n)$ . But by the inductive hypothesis,  $x = a_1 * \dots * a_p$  and  $y = a_{p+1} * \dots * a_n$  are independent of their bracketing. Hence we may rewrite them in the form:

$$\begin{aligned} x * y &= (a_1 * (a_2 * \dots * a_p)) * (a_{p+1} * \dots * a_n) \\ &= a_1 * ((a_2 * \dots * a_p) * (a_{p+1} * \dots * a_n)) \\ &\quad \text{(by associative law for three elements)} \\ &= a_1 * (a_2 * \dots * a_n) \\ &\quad \text{(by inductive hypothesis, bracketing of } n - 1 \text{ elements is unambiguous)} \end{aligned}$$

Hence all elements in  $S_n(a_1, \dots, a_n)$  are equal to  $a_1 * (a_2 * \dots * a_n)$ , and hence the set contains only one element. The generalised associative law follows.

### Properties of group tables

4. Explain why the identity element  $e$  must appear symmetrically about the leading diagonal in a group table.

*Solution:* If  $x * y = e$ , then  $y$  is the unique inverse of  $x$ . Therefore, we also have  $y * x = e$ . It follows that  $e$  is in both the  $(x, y)$  entry and the  $(y, x)$  entry, so the identity indeed must appear symmetrically in any group table.

5. Using the Latin square property of groups, show that there are at most  $(n - 1)!^{n-1}$  groups of order  $n$ , up to isomorphism. How tight is this bound for  $n = 1, 2, 3, 4$ ?

*Solution:* The first row and first column of a group table are fixed; they may be assumed to be the identity row and column without loss of generality. In the second row, we have a remaining  $n - 1$  elements which must be inserted. There are  $(n - 1)!$  ways of arranging these elements in the second row. We repeat this for all of the  $n - 1$  non-identity rows, implying a maximum of  $(n - 1)!^{n-1}$  groups of order  $n$  (note that this bound is not very tight at all, since once we have chosen one permutation in row two, we cannot use it again).

For  $n = 1$ , we have  $(0!)^0 = 1$ , so the bound is tight. For  $n = 2$ , we have  $(1!)^1 = 1$ , so again the bound is tight. For  $n = 3$ , we have  $(2!)^2 = 4$ , which is out by three (there is only one group of order three). Finally, for  $n = 4$ , we have  $(3!)^3 = 216$ , which is out by 214 (there are only two groups of order four).

Thus, this bound is pretty useless. *However*, this result does show something important: there are finitely many groups of any given finite order.

6. Are the following Latin squares group tables? Explain your answers.

| * | 1 | a | b | c | d |
|---|---|---|---|---|---|
| 1 | 1 | a | b | c | d |
| a | a | c | d | b | 1 |
| b | b | d | a | 1 | c |
| c | c | b | 1 | d | a |
| d | d | 1 | c | a | b |

| * | 1 | a | b | c | d |
|---|---|---|---|---|---|
| 1 | 1 | a | b | c | d |
| a | a | 1 | c | d | b |
| b | b | d | 1 | a | c |
| c | c | b | d | 1 | a |
| d | d | c | a | b | 1 |

*Solution:* In the first case, yes, since it is isomorphic to the group table of  $C_5$ . Swapping the second and third column, and the second and third rows, we have the table:

| * | 1 | a | c | b | d |
|---|---|---|---|---|---|
| 1 | 1 | a | c | b | d |
| a | a | c | b | d | 1 |
| c | c | b | d | 1 | a |
| b | b | d | 1 | a | c |
| d | d | 1 | a | c | b |

Now identifying  $1 \leftrightarrow e, a \leftrightarrow r, c \leftrightarrow r^2, b \leftrightarrow r^3, d \leftrightarrow r^4$ , the group table manifestly becomes the group table of  $C_5$ :

| *     | $e$   | $r$   | $r^3$ | $r^2$ | $r^4$ |
|-------|-------|-------|-------|-------|-------|
| $e$   | $e$   | $r$   | $r^3$ | $r^2$ | $r^4$ |
| $r$   | $r$   | $r^2$ | $r^3$ | $r^4$ | $e$   |
| $r^2$ | $r^2$ | $r^3$ | $r^4$ | $e$   | $r$   |
| $r^3$ | $r^3$ | $r^4$ | $e$   | $r$   | $r^2$ |
| $r^4$ | $r^4$ | $e$   | $r$   | $r^2$ | $r^3$ |

In the second case, no, since this is not associative. Note that according to the table, we have:

$$a * (b * d) = a * c = d, \quad (a * b) * d = c * d = a,$$

which are unequal.

### Power laws and periodicity

7. Find all the orders of the elements of: (a)  $C_4$ ; (b)  $D_3$ ; (c)  $Q_8$ ; (d) the chiral tetrahedral group.

*Solution:*

- (a) In  $C_4$ , the elements are  $\{e, r, r^2, r^3\}$ . Their orders are 1, 4, 2, 4 respectively.
- (b) In  $D_3$ , the elements are  $\{e, r, r^2, r^3, m, rm, r^2m, r^3m\}$ . Their orders are 1, 4, 2, 4, 2, 2, 2, 2 respectively (note all the reflections have order two).
- (c) In  $Q_8$ , the element orders are  $\{1, i, j, k, -1, -i, -j, -k\}$ . Their orders are  $\{1, 4, 4, 4, 2, 4, 4, 4\}$  respectively. Obviously  $(-1)^2 = 1$ , and with elements such as  $i$  we have  $i^2 = -1, i^3 = -i, i^4 = 1$ .
- (d) In the chiral tetrahedral group, there are three types of elements. The identity has order one. The rotations by  $2\pi/3$  or  $4\pi/3$  of the faces (i.e. about an axis through a vertex and the midpoint of an opposite face) clearly have order three. The rotations by  $\pi$  about axes through the midpoints of opposite edges have order two.

8. Let  $x, y$  be group elements. Show that: (a)  $x$  and  $x^{-1}$  have the same order; (b)  $x$  and  $xyx^{-1}$  have the same order; (c)  $xy$  and  $yx$  have the same order.

*Solution:*

- (a) Let  $x$  have order  $n$ . Now observe that  $(x^{-1})^k = (x^k)^{-1}$  by the power laws. Now if  $k \geq n$ , we have  $(x^{-1})^k = (x^k)^{-1} = e^{-1} = e$ . On the other hand, if  $k < n$  and  $(x^{-1})^k = e$ , this would imply  $(x^k)^{-1} = e$ , giving  $x^k = e$  after left-multiplying both sides by  $x^k$ , which is a contradiction. Thus  $k = n$  is the smallest integer for which  $(x^{-1})^k = e$ , and thus  $x^{-1}$  has order  $n$ .
- (b) Let  $x$  have order  $n$ . Then  $(xyx^{-1})^n = (xyx^{-1}) \dots (xyx^{-1}) = yx^n y^{-1}$  (cancelling all the intermediate pairs of  $y^{-1}y$ , and since  $x^n = e$ , we have  $yx^n y^{-1} = yy^{-1} = e$ . Therefore the order of  $xyx^{-1}$  is at most  $n$ .

On the other hand, suppose that  $xyx^{-1}$  has order  $m < n$ . Then  $e = (xyx^{-1})^m = (xyx^{-1})\dots(xy x^{-1}) = yx^m y^{-1}$ . Multiplying on the left by  $y^{-1}$  and the right by  $y$ , we have  $e = x^m$ , which contradicts the fact that  $x$  has order  $n$ . Thus  $xyx^{-1}$  has order exactly  $n$ .

(c) Let  $xy$  have order  $n$ . Then:

$$e = (xy)^n = x(yx)(yx)\dots(yx)y = x(yx)^{n-1}y.$$

Multiplying on the left by  $x^{-1}$  and on the right by  $y^{-1}$ , we have  $x^{-1}y^{-1} = (yx)^{n-1}$ . Recall that  $x^{-1}y^{-1} = (yx)^{-1}$ . So multiplying on both sides by  $yx$ , we have  $e = (yx)^n$ . Thus  $yx$  has order at most  $n$ . The argument works in reverse though, showing that if  $yx$  has order  $m$ , then  $xy$  has order at most  $m$ . So  $n \leq m$  and  $m \leq n$ , giving  $n = m$ .

9. Show that for any finite group  $G$ , there exists an integer  $n$ , such that  $x^n = e$  for all  $x \in G$ . The least such integer is called the *exponent* of the group. Calculate the exponents of  $C_6$  and  $D_3$ .

*Solution:* In a finite group  $G$ , every element has finite order. Let  $n_1, \dots, n_k$  be the orders of all the elements  $g_1, \dots, g_k$  of the group. Given any element  $g_i$  then, we have that:

$$g_i^{n_1 \dots n_k} = (g_i^{n_i})^{n_1 \dots n_{i-1} n_{i+1} \dots n_k} = e.$$

So if we take  $n = n_1 \dots n_k$ , then indeed for all  $g \in G$ , we have  $g^n = e$ .

In  $C_6 = \{e, r, r^2, r^3, r^4, r^5\}$  the orders of the elements are 1, 6, 3, 2, 3, 6 respectively. If we took  $n$  as above, we would take the product of all these elements. But notice that it is good enough to use the *lowest common multiple* of all the orders, since then for any element raised to the power, it will contain the order amongst that power. Therefore the exponent of  $C_6$  is six.

In  $D_3 = \{e, r, r^2, m, rm, r^2m\}$ , the orders of the elements are 1, 3, 3, 2, 2, 2 respectively. Again the exponent of  $D_3$  is six (which is the lowest common multiple of the orders).

10. Suppose that  $x, y$  are elements of an Abelian group, with orders  $n, m$  respectively. Is it true that the order of  $xy$  is the lowest common multiple of  $n, m$ ?

*Solution:* No, this is false. For example in  $C_2 = \{e, r\}$ , we have that  $r$  has order 2, but  $r^2 = e$  has order 1. The lowest common multiple of 2, 2 is 2.

11. Is it true that if all the elements of a group have finite order, then the group must be finite? Give a proof or counterexample as appropriate.

*Solution:* No, this is false, the group can be infinite. Consider the group of all rotations of the circle by an anticlockwise angle which is a rational multiple of  $2\pi$ . First, we verify that this is a group (let  $p, q, a, b$  be integers everywhere throughout, non-zero as appropriate):

(G1) If we rotate by  $2\pi p/q$ , then by  $2\pi a/b$ , we have rotated a total of  $2\pi(p/q + a/b)$ , which is still a rational multiple of  $2\pi$ .

(G2) Symmetry transformation composition is associative, as we discussed in the lecture. Alternatively, addition of rational numbers is associative!

(G3) Doing nothing is the same as rotating by an angle 0, which is a rational multiple of  $2\pi$ . Thus this set has an identity.

(G4) The inverse of a rotation by  $2\pi p/q$  is a rotation by  $-2\pi p/q$ , which is again a rational multiple of  $2\pi$  and hence included in the set.

All elements also have finite order. Note that repeating a rational rotation  $2\pi p/q$  a total of  $q$  times results in a rotation  $2\pi p$ , which is the same as doing nothing. Hence for each rotation in the group  $R$ , there is some integer  $m$  such that  $R^m = I$ . There must be a least such  $m$  in each case, implying that all elements have finite order. However, there are infinitely many rationals (between 0 and 1, crucially), providing the counterexample.

### Generators and relations

12. Show that  $\mathbb{Z}_6$  is generated by both  $\{1\}$  and  $\{2,3\}$ . What is the smallest size of a generating set for  $\mathbb{Z}_8^*$ ?

*Solution:* In the first case, note  $1 = 1, 2 = 1 + 1, 3 = 1 + 1 + 1, 4 = 1 + 1 + 1 + 1$ , etc. So  $\{1\}$  generates  $\mathbb{Z}_6$ .

In the second case, we have  $3 - 2 = 1$ , so we can generate 1 from this set. But then  $2 = 1 + 1 = (3 - 2) + (3 - 2)$ , so we must be able to generate everything else using the expressions we found above.

The group  $\mathbb{Z}_8^* = \{1, 3, 5, 7\}$  cannot be generated in terms of a single non-identity element, because it is not cyclic (we saw on the first Examples Sheet that it is isomorphic to  $D_2$ ). Therefore, the smallest size of a generating set is 2 (the same size as the generating set of  $D_2 = \{e, r, m, rm\}$  that we saw in the lecture).

13. Show that every group has a generating set. Is it true that every generating set of an infinite group is itself infinite? Give a proof or counterexample as appropriate.

*Solution:* Every group is generated by itself - just take the complete group, including all of its elements, as the generating set.

It is not true that an infinite group requires an infinite generating set; an easy counterexample is  $\mathbb{Z}$  under addition with generating set  $\{1\}$ .

14. A group  $G$  is generated by the distinct, non-identity elements  $a, b$  subject to the relations  $a^2 = b^2 = e$  and  $ab = ba$ . Construct the group table, and identify the group.

*Solution:* Since  $G$  is generated by  $a, b$ , a general element may be expressed in the form:

$$a^{\alpha_1} b^{\beta_1} a^{\alpha_2} b^{\beta_2} \dots a^{\alpha_k} b^{\beta_k}.$$

However, since  $ab = ba$ , everything commutes, so we may collapse this expression into something of the form:

$$a^{\alpha} b^{\beta}$$

by moving all  $a$ 's to the left and all  $b$ 's to the right. Note that  $a^2 = b^2 = e$ , so that  $\alpha = 0, 1$  and  $\beta = 0, 1$  are the only non-trivial choices. This gives a list of four possible elements:

$$\{e, a, b, ab\}.$$

We are given that  $e, a, b$  are all distinct. Further, if  $ab = e$ , then that would give  $a = b^{-1} = b$ , which is a contradiction. If  $ab = a$ , we would have  $b = e$  which is a contradiction. Similarly, if  $b = ab$ , we would have  $a = e$ , which is a contradiction. Thus  $ab$  must be genuinely distinct element.

We now have enough information to construct the group table:

| $\cdot$ | $e$  | $a$  | $b$  | $ab$ |
|---------|------|------|------|------|
| $e$     | $e$  | $a$  | $b$  | $ab$ |
| $a$     | $a$  | $e$  | $ab$ | $b$  |
| $b$     | $b$  | $ab$ | $e$  | $a$  |
| $ab$    | $ab$ | $b$  | $a$  | $e$  |

15. Let  $r, m$  be the generators of  $D_n$  defined in this chapter. Show by induction that:

$$(r^\alpha m^\beta)^k = r^{\alpha S_k} m^{k\beta},$$

where  $S_k = 1 + (-1)^\beta + (-1)^{2\beta} + \dots + (-1)^{(k-1)\beta}$ . Deduce that  $D_n$  is not a cyclic group.

*Solution:* The base case is  $k = 1$ , which is clear. Assume the result is true for general  $k$ , and consider the case with  $k + 1$ . We have:

$$(r^\alpha m^\beta)^{k+1} = (r^\alpha m^\beta)(r^{\alpha S_k} m^{k\beta}).$$

Using the rule for moving rotations past reflections, we have:

$$r^{\alpha + \alpha S_k (-1)^\beta} m^{(k+1)\beta}.$$

Observe that  $S_{k+1} = 1 + S_k(-1)^\beta = 1 + (-1)^\beta + (-1)^{2\beta} + \dots + (-1)^{k\beta}$ , and thus the result is proved by induction.

Now if  $D_n$  were a cyclic group, it would be generated by one element  $r^\alpha m^\beta$ . If that element were to have  $\beta = 0$ , it is clear from the above that we would only generate rotations. On the other hand, if that element were to have  $\beta = 1$ , then  $S_k = 1$  for  $k$  odd, and  $S_k = 0$  for  $k$  even. This gives the list of elements generated by  $r^\alpha m$  as:

$$e, \quad r^\alpha m, \quad e, \quad r^\alpha m, \quad \text{etc.}$$

This is only two elements! So we have not generated all of  $D_n$ , and hence it is not a cyclic group.

16. Suppose that all the non-identity elements of a group have order two.

- (a) Show that the group must be Abelian.
- (b) Show that, if the group is finite, then it has order  $2^n$  for some  $n$ . [Hint: consider a minimal generating set.] Can the group be infinite?

*Solution:*

- (a) Let  $x, y$  be two distinct elements of the group, and consider  $xy$ . This element must either be the identity, or must have order two. If it is the identity, then  $xy = e$  implies  $x = y^{-1} = y$ , since  $y$  has order two; this contradicts our assumption that the elements were distinct. In the remaining case,  $xy$  must have order two:

$$(xy)^2 = e \quad \Rightarrow \quad xyxy = e \quad \Rightarrow \quad yx = x^{-1}y^{-1}.$$

Using  $x^{-1} = x, y^{-1} = y$ , we deduce that  $yx = xy$ , implying that the group is Abelian.

- (b) Let  $\{x_1, \dots, x_k\}$  be a generating set for the group of smallest possible size; such a set obviously excludes the identity, as this does not help us generate anything. Then, since the group is Abelian, a general element may be expressed in the form:

$$x_1^{\alpha_1} \dots x_k^{\alpha_k}.$$

Here, the only non-trivial choices of the exponents are  $\alpha_i = 0, 1$ , since  $x_i$  has order two (it is not the identity). Now suppose that two of these expressions are the same:

$$x_1^{\alpha_1} \dots x_k^{\alpha_k} = x_1^{\beta_1} \dots x_k^{\beta_k} \quad \Rightarrow \quad x_1^{\alpha_1 - \beta_1} \dots x_k^{\alpha_k - \beta_k} = e.$$

If, without loss of generality, we have  $\alpha_1 \neq \beta_1$ , then we see that we can write:

$$x_1 = x_2^{\gamma_1} \dots x_k^{\gamma_k},$$

for some  $\gamma_i$ . This contradicts the minimality of the generating set. Thus all expressions of the given form are unique, giving a total of  $2^k$  elements.

The group *can* be infinite. The set of all possible sequences  $(x_1, x_2, x_3, \dots)$  where  $x_i$  are taken modulo 1, under addition, gives an example of such a group.